

Insider-Buying ETF Signal Strategy

A meticulous, point-in-time framework for converting SEC Form 4 purchases into stock conviction signals, ETF discovery, and multi-strategy portfolio evidence

PRIMARY DATA SEC Form 4 / 4-A	IMPLEMENTATION Stock-first ETF discovery	TARGET ROLE Alpha + conviction sensor
----------------------------------	---	--

Core research question

Do clusters of recent, disclosed open-market purchases by corporate officers and directors identify stock groups whose common ETF exposures subsequently earn incremental risk-adjusted returns?

Prepared for a modular trading-agent architecture with independent alpha engines, a portfolio allocator, and a separate hedge overlay.

Executive Summary

This document converts the insider-buying idea into a falsifiable, research-grade strategy. The strategy begins with publicly disclosed insider purchases at individual companies, scores the strength and freshness of those purchases, discovers ETFs that hold the qualifying stocks, and ranks the ETFs by the concentration and breadth of insider conviction embedded in their holdings.

Recommended data architecture

Use the SEC Insider Transactions Data Sets as the free historical backbone; enrich each accession with EDGAR filing-time metadata and full XML when needed; use QuantConnect for point-in-time ETF constituents, security mapping, prices, fees, and execution. FMP is optional as a convenience feed or cross-check, not a required source.

Component	Canonical recommendation
Economic thesis	Insiders committing personal capital may possess useful information or superior valuation judgment; multiple independent purchases are potentially more informative than one isolated trade.
V1 transaction	Form 4 non-derivative common-stock transaction with code P, acquired indicator A, valid shares and price, officer or director, purchase value at least \$50,000.
Signal clock	Use the public filing availability time, never the transaction date. When only the filing date is known, enter no earlier than the next trading-day open.
Stock signal	Purchase value scaled logarithmically, decayed with a 20-trading-day half-life, aggregated over a 30-day lookback. Cluster count is retained as a separate feature.
ETF discovery	Build a point-in-time reverse index from stock to eligible ETFs. Candidate ETFs are the union of ETFs holding one or more qualifying stock seeds.
ETF score	Primary: sum of ETF constituent weight multiplied by standardized stock score. Also retain seed count, weighted seed exposure, breadth, coverage, and concentration.
Positioning	Initially long-only: overweight the highest-scoring ETFs. Do not mirror insider sales into a symmetric short signal in V1.
Research gate	Validate stock-level event information first, then ETF aggregation, then investable portfolio performance, and only afterward allocator and hedge integration.

The likely value of the signal is not a standalone “buy every insider purchase” strategy. Its stronger role is as an independent evidence channel inside a multi-strategy system: analysts express what informed outsiders believe, while insiders reveal where corporate decision-makers are committing their own capital. When this signal agrees with earnings, momentum, short-interest, or fundamental-acceleration engines, the portfolio allocator can treat the agreement as higher-confidence evidence.

Decision Summary

- Proceed with the research: the data is obtainable at low cost, the hypothesis is economically coherent, and the signal is sufficiently different from analyst ratings to improve diversification.
- Start with purchases only. Insider sales are noisier because liquidity, taxation, diversification, compensation, estate planning, and prearranged plans can motivate sales without a negative view.
- Keep the first model simple. Test transaction value, freshness, and clustering separately before introducing subjective role weights, drawdown multipliers, machine learning, or leverage.
- Treat point-in-time engineering as the first milestone. A profitable result is invalid if amendments, filing availability, symbol changes, joint filers, or ETF holdings lags are mishandled.

Contents

Section	Topic
Part I	Thesis, scope, and transaction meaning
1	Strategy thesis and expected role
2	What counts as an insider-buying signal
3	Two-level clustering hypothesis

Section	Topic
Part II	Data and point-in-time engineering
4	Data-source recommendation
5	SEC bulk data ingestion
6	EDGAR live and timestamp enrichment
7	Normalized schemas and security master
8	Hard cases, amendments, and footnotes
Part III	Signal construction
9	Canonical stock-level signal
10	Feature extensions and ablations
11	Seed-stock selection
Part IV	ETF discovery and portfolio use
12	Stock-to-ETF reverse index
13	ETF eligibility and scoring
14	Portfolio construction and allocator contract
Part V	Research execution
15	Backtest sequence and controls
16	Statistics, robustness, and stage gates
Part VI	Implementation blueprint
17	Collection jobs, storage, and code layout
18	Worked example and pseudocode
19	Go/no-go criteria and risk register
Appendices	Form codes, field dictionary, experiment IDs, references

PART I

Thesis, Scope, and Transaction Meaning

Define the investment claim before building the pipeline.



1. Strategy Thesis and Expected Role

The strategy is a stock-first, event-driven ETF selection system. It does not begin by screening every ETF. It begins with disclosed purchases by corporate insiders, converts those events into stock-level conviction, then searches for ETFs in which several qualifying stocks cluster with meaningful portfolio weight.

Insider purchase events -> Stock conviction -> Stock-to-ETF lookup -> ETF conviction -> Portfolio allocator

The hedge layer remains separate and operates only after strategy positions are netted.

1.1 Economic intuition

A qualifying purchase can contain information for at least three non-exclusive reasons. First, the insider may have superior knowledge of operating conditions. Second, the insider may understand the company's long-term value better than outside investors. Third, the purchase may communicate confidence to the market even when no private information is involved. The strategy does not need to identify which channel dominates; it needs the disclosed purchase pattern to predict subsequent returns after controlling for known confounders.

The ETF layer changes the wager. Buying one company after one insider purchase leaves the portfolio exposed to company-specific error. Buying an ETF only when purchases cluster across multiple constituents attempts to preserve the common industry or theme component while reducing idiosyncratic risk.

1.2 Expected strategic role

Role	Interpretation
Standalone alpha	Possible but not the base-case expectation. The signal must survive costs and factor controls before being treated as a standalone strategy.
Cross-sectional selector	Primary use: rank otherwise eligible ETFs and overweight the strongest insider-conviction clusters.
Confirmation sensor	High value when independent engines such as earnings, analyst revisions, momentum, or fundamentals agree.
Risk filter	Useful for refusing or reducing a position when other engines are bullish but insider evidence is absent or contradictory; absence alone is not a bearish signal.
Hedge trigger	Not recommended. The hedge engine should respond to total portfolio exposure and regime risk, not directly to this one signal.

1.3 Research hypotheses

ID	Pre-registered hypothesis
H1	Qualifying code-P purchases predict positive future stock excess returns after public filing.
H2	Multiple unique insiders buying the same issuer within a short window are more informative than one isolated purchase.
H3	ETFs containing multiple qualifying issuer signals outperform ETFs with weaker or narrower insider evidence.
H4	ETF-weighted stock conviction adds information beyond simple seed-stock count.
H5	The ETF signal remains incremental after controlling for size, value, momentum, prior drawdown, earnings events, liquidity, sector, and market regime.
H6	The positive signal is asymmetric: purchases are more reliable than sales as a directional input.

2. What Counts as an Insider-Buying Signal

The term “insider acquisition” is dangerously broad. Form 4 reports grants, option exercises, tax withholding, gifts, conversions, purchases, and sales. The first model must isolate events that plausibly represent a voluntary commitment of capital.

2.1 Canonical V1 inclusion filter

1. Filing is Form 4 or a reconciled Form 4/A amendment.

2. Transaction appears in the non-derivative transaction table and represents common equity or an equivalent ordinary share class.
3. Transaction code is P, which the SEC defines as an open-market or private purchase.
4. Acquired/disposed indicator is A.
5. Shares and price per share are positive, parseable, and not solely footnote ranges that cannot be resolved.
6. Reporting owner is an officer or director. Ten-percent owners are excluded in the canonical model unless they are also an officer or director.
7. Purchase value is at least \$50,000 after aggregation rules are applied.
8. Signal is timestamped by public filing availability, not by transaction date.

Why code P only?

Code A is generally a grant or award; M is an exercise or conversion; F is tax or exercise-price withholding; G is a gift; and J is an "other" transaction requiring footnote interpretation. These events may be valuable research features later, but mixing them into V1 would dilute the clean personal-capital thesis.

Code	SEC meaning (abridged)	V1 treatment
P	Open-market or private purchase	Include in V1
S	Open-market or private sale	Exclude; future separate research
A	Grant, award, or other Rule 16b-3 acquisition	Exclude
M	Exercise or conversion of derivative security	Exclude
F	Shares withheld/delivered for taxes or exercise price	Exclude
G	Gift	Exclude
J	Other acquisition or disposition	Quarantine / footnote review
L	Small acquisition under Rule 16a-6	Exclude from canonical test

2.2 Transaction date versus signal date

The transaction date tells us when the insider acted. The filing acceptance time tells us when the market could know. The backtest must use the latter. Form 4 is generally due within two business days, so using the transaction date creates up to two days of look-ahead bias and can materially exaggerate performance.

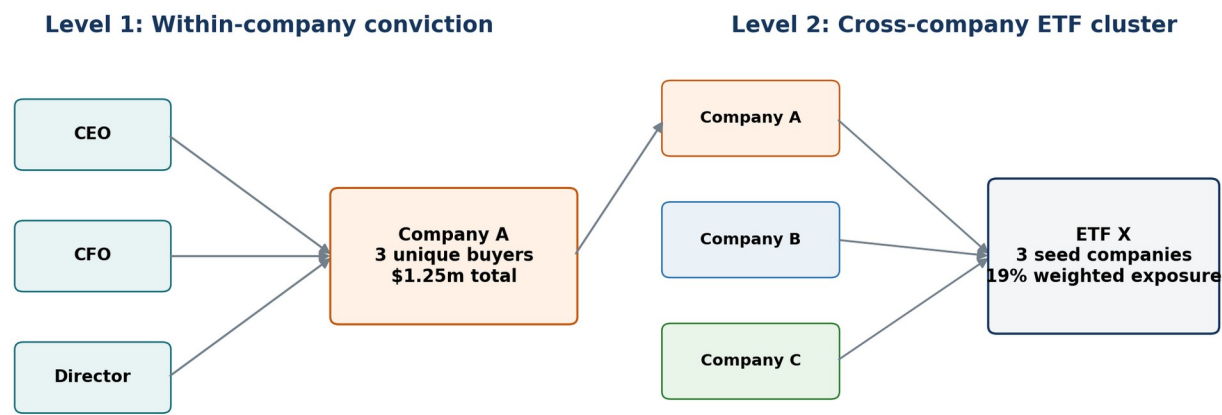
```
signal_available_time = EDGAR acceptance time; earliest_trade_time > signal_available_time
```

If only FILING_DATE is available, the conservative primary rule is the next trading-day open.

2.3 Purchases are not all equally discretionary

- Open-market purchase: usually the cleanest expression of discretionary capital commitment.
- Private purchase: code P can also include a private transaction; retain it but flag it, cap extreme values, and run a robustness test excluding likely strategic/private placements.
- Direct ownership: easier to interpret and to calculate ownership increase. Use as the primary sample.
- Indirect ownership through trusts, spouses, partnerships, or entities: retain as a separate sample because control and economic exposure may differ.
- Rule 10b5-1 indication: retain as a feature. A prearranged purchase may be less responsive to current information but can still communicate long-term conviction. Historical coverage is limited, so do not use it as a universal filter.

3. The Two-Level Clustering Hypothesis



The distinctive hypothesis is not merely "an insider bought". It is that multiple independent insider decisions cluster first within issuers and then across companies connected through an investable ETF.

The strongest version of the strategy has two distinct clustering layers. First, several insiders may independently buy the same company. Second, several companies experiencing insider buying may cluster inside the same ETF. These layers should be measured separately so the research can determine whether each adds incremental information.

Layer	Measure	Interpretation
Within-company cluster	Unique officers/directors buying one issuer during a window	Captures internal agreement and reduces dependence on one person.
Cross-company ETF cluster	Unique qualifying issuers represented in one ETF	Captures common sector/theme information while diversifying company-specific error.
Weighted cluster	ETF weight multiplied by stock conviction	Distinguishes a meaningful exposure from a token holding.
Institutional breadth	Different insider roles, ownership types, and transaction dates	Diagnostic only at first; avoid subjective weights before evidence.

PART II

Data and Point-in-Time Engineering

Build the evidence ledger before testing the strategy.



4. Data-Source Recommendation

The best cost-conscious architecture uses public SEC data for the insider layer and QuantConnect for market/ETF infrastructure. A paid normalized feed can accelerate live operation, but it should not be the only audit trail.

Source	Role	Strength	Recommendation
SEC Insider Transactions Data Sets	Primary historical backfill	Free quarterly flattened Form 3/4/5 data; January 2006 through the current posted quarter; as-filed tables and accession numbers.	Best historical backbone
SEC EDGAR filing archives / feeds	Timestamp and XML enrichment; live incremental	Exact filing artifacts, acceptance metadata, footnotes, amendments, and auditability.	Required for high-integrity PIT
Financial Modeling Prep	Optional convenience feed	Normalized latest/search APIs with dates, roles, transaction details, and filing links.	Useful, not mandatory
QuantConnect	ETF holdings, prices, security master, execution	Historical ETF constituents and weights, security mapping, corporate actions, fee/slippage simulation.	Primary ETF/backtest platform

Important discovery

The SEC now publishes a free, flattened Insider Transactions Data Set covering XML submissions from January 2006 through the current posted period. As of August 2026, the page lists data through June 2026 and updates quarterly. This materially lowers the cost of the research.

4.1 Recommended historical window

- Insider source availability: January 2006 onward through SEC bulk data.
- QuantConnect ETF constituent availability: June 2009 onward; monthly before January 2015 and daily thereafter.
- Canonical primary test: January 2015 onward, because daily ETF holdings reduce interpolation and stale-composition risk.
- Secondary long-history test: June 2009-December 2014 using monthly constituent snapshots and lower rebalance frequency.
- Final holdout: reserve the most recent untouched period. If 2024-2026 has already influenced design decisions, select another genuinely unexamined holdout.

5. SEC Bulk Data Ingestion

Each quarterly SEC package contains up to eight tab-delimited UTF-8 files. The core joins are `ACCESSION_NUMBER` across `SUBMISSION`, `REPORTINGOWNER`, and `NONDERIV_TRANS`. `FOOTNOTES` and `OWNER_SIGNATURE` are used for audit and exceptional parsing. The raw files must be retained exactly as downloaded with checksums.

File	Purpose	Key information
SUBMISSION	Filing identity and issuer	ACCESSION_NUMBER, FILING_DATE, PERIOD_OF_REPORT, DOCUMENT_TYPE, ISSUERCIK, ISSUENAME, ISSUERTRADINGSYMBOL, remarks and later 10b5-1 metadata.
REPORTINGOWNER	Who filed and relationship	RPTOWNERCIK, name, officer/director/10% owner/other relationship, title.
NONDERIV_TRANS	Canonical V1 transactions	Security title, transaction date, code, shares, price, acquired/disposed, shares owned after, direct/indirect ownership.
NONDERIV_HOLDING	Post-transaction holdings	Useful for ownership reconciliation and Form 3/5 context.
DERIV_TRANS / HOLDING	Options and derivative activity	Exclude from V1; retain for future research and reconciliation.
FOOTNOTES	Transaction qualifiers	Needed for price ranges, ownership structures, private transactions, and "other" codes.
OWNER_SIGNATURE	Signature identity	Audit and entity-resolution support.

5.1 Raw-storage rules

```
data/  
  raw/sec/insider_bulk/  
    2006q1_form345.zip  
    ...  
    2026q2_form345.zip
```

```

manifests/
  sec_insider_bulk_manifest.jsonl
normalized/
  insider_filings.parquet
  insider_owners.parquet
  insider_transactions.parquet
derived/
  insider_stock_signals.parquet
  stock_etf_reverse_index.parquet
  insider_etf_signals.parquet

```

- Never overwrite a raw quarterly ZIP or parsed as-filed table.
- Store SHA-256, source URL, retrieval timestamp, byte size, parser version, and Git commit in a manifest.
- Partition normalized data by filing year/quarter, not by current ticker.
- Retain accession number as the immutable filing identifier and transaction surrogate key as the row identifier.
- Persist both raw fields and normalized fields. A parser update must not destroy the original interpretation context.

5.2 Historical batch process

9. Download every quarterly package from 2006 through the latest available quarter.
10. Validate ZIP integrity and expected file inventory.
11. Load tab-delimited files with explicit schemas; do not rely on type inference for identifiers.
12. Create filing, owner, non-derivative transaction, derivative transaction, holdings, footnote, and signature tables.
13. Identify Form 4 and Form 4/A records; retain Forms 3 and 5 for ownership context and missing-report diagnostics.
14. Join non-derivative transactions to issuer and owner metadata without multiplying transaction rows by joint-owner rows.
15. Classify security titles and transaction codes; quarantine unresolved records.
16. Enrich with EDGAR acceptance time and original XML where needed.
17. Version the normalized snapshot before any signal features are created.

6. EDGAR Live and Timestamp Enrichment

The quarterly bulk package is excellent for backfill, but it omits certain metadata and is not a live feed. Each accession used in the canonical research sample should be linked back to the filing archive. This enables acceptance-time recovery, full XML parsing, amendment handling, and human audit.

6.1 Availability timestamp hierarchy

Tier	Available metadata	Execution rule
Tier A	Exact EDGAR acceptance timestamp	Trade only after acceptance; next open is the conservative default.
Tier B	Filing date plus known market session	Apply session-specific lag; never use a price before the disclosure.
Tier C	Filing date only	Primary rule: next trading-day open. Do not infer morning availability.
Invalid	Transaction date only	Not point-in-time for a public-following strategy; exclude or enrich.

6.2 Fair-access implementation

- Use an identifying User-Agent with a monitored contact address.
- Respect the SEC's current fair-access limit of no more than 10 requests per second across all machines; use a lower internal target such as 5 requests per second.
- Cache every filing artifact by accession number and never redownload unchanged files unnecessarily.
- Use exponential backoff, request checkpoints, and resumable jobs.
- Do not run thousands of filing requests inside a backtest. Enrich and freeze the dataset beforehand.

6.3 Live incremental feed

For paper or live operation, poll or consume current EDGAR filing indexes for Forms 4 and 4/A, fetch the ownership XML, normalize the event, apply the same V1 filters, and append the immutable event to the live evidence ledger. A vendor API such as FMP can serve as a low-latency convenience feed, but the accession-linked SEC filing should remain the authoritative audit object.

7. Normalized Schemas and Security Master

Tickers are display labels, not durable identifiers. The normalized layer must join SEC issuer CIK, QuantConnect symbol/security identifiers, vendor symbols, and historical ticker ranges into one security master.

Field	Meaning	Use
security_id	Internal immutable identifier	Primary join key across strategies
issuer_cik	SEC issuer Central Index Key	Primary SEC entity key
qc_symbol_id	QuantConnect mapped security identifier	Market data and corporate actions
ticker	Ticker valid for a date range	Display and vendor requests only
valid_from / valid_to	Ticker/security mapping interval	Prevents future symbol mappings
share_class	Common A/B, ADR, preferred, etc.	Controls what qualifies as ordinary equity
exchange / country	Listing metadata	Universe eligibility

7.1 Canonical normalized transaction table

```

insider_transaction_id
accession_number
filing_date
acceptance_timestamp
transaction_date
issuer_cik
security_id
security_title_raw
security_class_normalized
reporting_owner_cik
reporting_owner_name
is_officer
is_director
is_ten_percent_owner
officer_title_raw
role_normalized
transaction_code
acquired_disposed_code
shares
price_per_share
purchase_value
direct_indirect
shares_owned_after
shares_owned_before_estimate
ownership_increase_pct
is_10b5_1
is_amendment
original_accession_number
footnote_ids
source_snapshot_version

```

7.2 Purchase-value and ownership calculations

$$\text{PurchaseValue}_e = \text{Shares}_e * \text{PricePerShare}_e$$

$$\text{SharesOwnedBefore}_e = \text{SharesOwnedAfter}_e - \text{SharesAcquired}_e$$

Use only when the same security, ownership type, and transaction sequence make this identity valid.

$$\text{OwnershipIncrease}_e = \text{SharesAcquired}_e / \max(\text{SharesOwnedBefore}_e, \text{epsilon})$$

Ownership-increase calculations are fragile when the filing reports multiple transactions, indirect holdings, jointly owned securities, or footnote adjustments. Treat the value as high-confidence only when reconciliation checks pass; otherwise leave it missing rather than inventing precision.

8. Hard Cases, Amendments, and Footnotes

Case	Risk	Required handling
Form 4/A	Correction can change date, price, shares, owner, or code	Version filings; reconcile to original; supersede corrected rows rather than counting both.
Joint filing	Several reporting owners can share one filing	Do not multiply one transaction by owner count. Attach owner set; count unique buyers only when the transaction is economically attributable to each.
Multiple transactions	One filing may contain several purchase lots	Aggregate same owner/security/date after preserving lot-level lineage.
Price range in footnote	Price field may be average or blank	Parse footnote if deterministic; otherwise quarantine from value-based V1 and retain for count-only tests.
Indirect ownership	Trust, spouse, partnership, LLC, or entity	Primary sample direct only; secondary sample flags indirect transactions.
Private purchase	Code P includes private purchases	Retain flag; winsorize value and run exclusion robustness.
10% owner	Can be a fund or strategic blockholder, not management	Separate sample; exclude from canonical officer/director thesis.
Form 5	Delayed annual reporting	Exclude from canonical timely signal; use only to assess missing small transactions.
Security title ambiguity	Restricted stock, phantom units, preferred, ADR, class variants	Deterministic title mapping plus manual exception dictionary.

Fail closed, not open

A transaction that cannot be confidently classified should not silently enter the alpha sample. Send it to a quarantine table with a reason code. Coverage loss is preferable to contamination.

PART III

Signal Construction

Begin with a transparent baseline; add complexity only through ablations.



9. Canonical Stock-Level Signal

The first serious specification should contain as few discretionary choices as possible. Purchase size and freshness have direct interpretations; role weights, drawdown interactions, rarity, and insider historical quality are useful extensions but should not enter V1.

9.1 Event score

$$\text{Size}_e = \ln(1 + \text{PurchaseValue}_e / 50,000)$$

$$\text{Freshness}_e(t) = \exp[-\ln(2) * \text{AgeTradingDays}_e(t) / 20]$$

$$\text{EventScore}_e(t) = \text{Size}_e * \text{Freshness}_e(t)$$

The logarithm prevents a \$20 million purchase from mechanically overwhelming dozens of meaningful \$100,000-\$1 million purchases. The 20-trading-day half-life is a pre-registered starting point, not an optimized truth. Separate experiments should test 5, 10, 20, 40, and 60 trading days.

9.2 Stock score

$$\text{RawStockScore}_i(t) = \text{sum of EventScore}_e(t) \text{ for qualifying events in issuer } i \text{ during the last 30 trading days}$$

$$\text{StockScore}_i(t) = \text{winsorized cross-sectional z-score of RawStockScore}_i(t)$$

Calculate additional fields without combining them into the primary score: unique buyers, CEO-buy flag, CFO-buy flag, total purchase value, direct-purchase share, ownership-increase median, days since previous insider purchase, and whether the purchase occurred under a 10b5-1 plan.

9.3 Two distinct breadth measures

Feature	Definition	Interpretation
Buyer breadth	Number of unique qualifying insiders buying one issuer in the lookback	Within-company agreement
Role breadth	Number of distinct normalized roles represented	CEO/CFO/director participation
Date breadth	Number of distinct purchase dates	Separates one joint event from repeated conviction
Dollar breadth	Share of total value not attributable to the largest buyer	Reduces one-buyer concentration

10. Feature Extensions and Ablations

The research should add one feature family at a time. Each extension receives a new experiment identifier and must improve out-of-sample performance or diagnostics before it is retained.

Family	Examples	Research treatment
Role	CEO, CFO, other officer, director	Test one-hot features first; avoid arbitrary multiplicative weights.
Ownership increase	Shares purchased relative to prior holdings	Potentially stronger conviction than dollar value alone; missingness must be handled explicitly.
Cluster	Unique buyers and repeated purchase dates	Core extension; likely one of the most informative features.
Rarity	Days since prior purchase by insider/issuer	First purchase in years may be more meaningful; test separately.

Family	Examples	Research treatment
Price context	Drawdown, distance from 52-week high, momentum	Separates value-buying from trend confirmation; do not assume which is superior.
Valuation	Book-to-market, earnings yield, FCF yield	Controls whether insider buying merely proxies value.
Liquidity/size	Market cap, ADV, spread, idiosyncratic volatility	Essential confounders and implementability filters.
10b5-1	Affirmative-defense plan indicator where available	Potential discretion classifier; limited historical coverage.
Insider history	Past event alpha by person	Advanced; must be recursively estimated with shrinkage.
Opportunistic/routine	Pattern-based trade classification	Research-supported concept; implement only after transparent baselines.

10.1 Suggested ablation ladder

INSETF-S1-VALUE
purchase value only

INSETF-S2-FRESHNESS
value * freshness

INSETF-S3-CLUSTER
S2 + unique-buyer features

INSETF-S4-OWNERSHIP
S3 + ownership increase

INSETF-S5-ROLE
S4 + role one-hot features

INSETF-S6-CONTEXT
S5 + drawdown / momentum / valuation controls

INSETF-S7-OPPORTUNISTIC
pattern-based routine-versus-opportunistic classifier

11. Seed-Stock Selection

Seed stocks are the bridge between the insider event layer and ETF discovery. A stock becomes a seed because its point-in-time insider score exceeds a frozen rule; it must not be selected because its subsequent return was attractive.

Method	Rule	Trade-off
Threshold seed	Raw score above a fixed economic threshold	Easy to interpret; population varies through time.
Percentile seed	Top 5% or top 10% of eligible stocks each date	Stable breadth; can admit weak signals during quiet periods.
Cluster-gated seed	At least 2 unique insiders and minimum value	Strong thesis alignment; smaller sample.
Hybrid seed	Economic minimum plus percentile rank	Recommended primary design.

Canonical seed rule

A stock qualifies when it has at least one V1 purchase of \$50,000 or more, a positive 30-day decayed stock score, and ranks in the top 10% of eligible stocks on the signal date. Run a separate cluster-gated test requiring at least two unique buyers.

11.1 Eligibility before scoring

- U.S.-listed common equity supported by the market-data/security-master layer.

- Price at least \$5 and 20-day average dollar volume at least \$2 million for stock-event tests; stricter thresholds may be used for direct stock trading but ETF implementation is less sensitive.
- At least 60 trading days of price history for context controls.
- No known stale/delisted mapping on the signal date.
- No unresolved amendment or transaction-classification conflict.

11.2 Seed-selection diagnostics

Diagnostic	Required output
Eligible-stock count	Number of securities that could have become seeds before observing the signal.
Qualifying event count	Code-P purchase events surviving security, role, value, amendment, and availability filters.
Seed count and percentile	Selected population, frozen threshold, and cross-sectional percentile on each date.
Score concentration	Share of aggregate seed evidence generated by the top one, five, and ten stocks.
Buyer breadth	Unique buyers and issuers; separates many filings from many independent decisions.
Sector and size mix	Seed distribution by sector, market-cap bucket, liquidity, and prior drawdown.
Entry/exit turnover	How often securities enter or leave the seed set as freshness decays.

The seed rule is part of the model

Record the entire eligible population and the exact rejection reason for every non-seed. A reverse lookup that begins only from remembered winners cannot be audited for selection bias.

PART IV

ETF Discovery and Portfolio Use

Invert historical holdings, score candidate ETFs, and output evidence - not uncontrolled leverage.



12. Stock-to-ETF Reverse Index

QuantConnect's interface is naturally ETF-to-constituents. The stock-first strategy therefore requires a reverse index built from an independently defined, point-in-time eligible ETF universe. This index should be precomputed rather than reconstructed inside every backtest event.

$$\text{ReverseIndex}(t, \text{stock}_i) = [(\text{ETF}_j, \text{constituent_weight}_{i,j,t}), \dots]$$

```
for date in trading_dates:
    for etf in eligible_etfs(date):
        snapshot = point_in_time_constituents(etf, date)
        for constituent in snapshot:
            write(date, constituent.security_id, etf.security_id, constituent.weight)

# Query at signal time
candidate_etfs = union(reverse_index[date][stock] for stock in seed_stocks)
```

12.1 Point-in-time holdings rule

The ETF snapshot used at date t must have been observable by t . QuantConnect notes that its constituent data can be delayed by up to one week. The implementation must inspect available date attributes such as `EndTime` and `LastUpdate`. If availability semantics cannot be proven, lag holdings by five trading days in the primary test and treat unlagged results as exploratory.

12.2 Why the eligible ETF universe remains independent

Stock-first discovery does not eliminate the need for an ETF universe. On every historical date, the strategy must intersect ETFs that hold the seed stocks with ETFs that actually existed, were tradable, met liquidity rules, and were not leveraged or inverse. Otherwise, current-survivor and signal-conditioned selection biases can enter silently.

$$\text{CandidateETFs}_t = \text{ETFsHoldingSeeds}_t \text{ INTERSECT } \text{EligibleETFs}_t$$

13. ETF Eligibility and Scoring

13.1 Canonical ETF eligibility

Filter	Canonical rule
Instrument type	U.S.-listed long equity ETF; exclude leveraged, inverse, volatility, commodity, bond, crypto, and closed-end products.
History	At least 252 trading days since inception.
Price	At least \$5.
Liquidity	20-day median dollar volume at least \$5 million.
Holdings quality	Known constituent weights covering at least 90% of the fund or a documented residual/cash bucket.
Seed breadth	At least two qualifying seed stocks for the primary ETF-cluster portfolio; one-seed ETFs remain a diagnostic sample.
Seed exposure	At least 5% of ETF weight in qualifying seeds, unless a preregistered alternative threshold is tested.

13.2 Score family

Metric	Formula/definition	Purpose
SeedCount	Number of qualifying seed stocks held	Original intuitive signal; mandatory baseline.
SeedExposure	Sum of ETF weights of qualifying seeds	Measures how much of the fund is touched by the signal.
ConvictionScore	Sum(weight * standardized stock score)	Primary weighted alpha score.

Metric	Formula/definition	Purpose
BuyerBreadth	Weighted or unweighted unique insider counts across seed issuers	Captures evidence density.
Coverage	ETF weight with valid insider-data eligibility	Reliability diagnostic; not every holding is equally observable.
Concentration	Largest stock contribution / total score	Flags one-stock domination.
ClusterOverlap	Similarity to other high-scoring ETFs	Prevents treating near-duplicate ETFs as independent bets.

$$\text{ETFConviction}_j(t) = \text{sum}_i [\text{Weight}_{i,j,t} * \text{StockScore}_i(t)]$$

$$\text{ETFSeedExposure}_j(t) = \text{sum}_i [\text{Weight}_{i,j,t} * I(\text{Stock}_i \text{ is a seed})]$$

$$\text{ETFConcentration}_j(t) = \text{max}_i \text{Contribution}_{i,j,t} / \text{sum}_i \text{Contribution}_{i,j,t}$$

Do not collapse the diagnostics too early

Backtest seed count, weighted exposure, conviction score, breadth, and concentration separately before inventing one composite formula. A simple count may outperform a beautiful equation; the research must be allowed to say so.

13.3 Duplicate ETF exposure

ETFs such as sector peers can hold substantially the same securities. Calculate holdings overlap or cosine similarity and cluster near-duplicates. Run results both with all eligible ETFs and with one representative per cluster. Portfolio allocation should cap exposure to a holdings cluster even when several ETFs rank highly.

$$\text{HoldingsOverlap}(a,b) = \text{sum}_i \min(\text{Weight}_{i,a}, \text{Weight}_{i,b})$$

14. Portfolio Construction and Allocator Contract

This alpha engine should output a standardized evidence object. The portfolio allocator - not the signal engine - decides position size after considering other strategies, correlation, net sector exposure, volatility, and the hedge overlay.

14.1 Initial investable portfolio

- Rebalance weekly using the latest valid evidence and known ETF holdings.
- Select the top 3-5 eligible ETFs by conviction score, subject to holdings-cluster and sector caps.
- Equal-weight the selected ETFs in the first portfolio test; compare against score weighting and volatility scaling later.
- Maximum 25% per ETF, 40% per sector/theme, and 35% per ETF-overlap cluster.
- Hold cash when fewer than the minimum number of ETFs pass the signal threshold.
- No leverage in the alpha-validation phase.

14.2 Strategy output contract

```
{
  "strategy_id": "insider_buying_etf",
  "as_of": "2026-08-21T13:00:00-04:00",
  "instrument": "ETF_X",
  "direction": 1,
  "score": 0.83,
  "confidence": 0.71,
  "horizon_trading_days": 20,
  "seed_count": 4,
  "seed_exposure": 0.19,
```

```

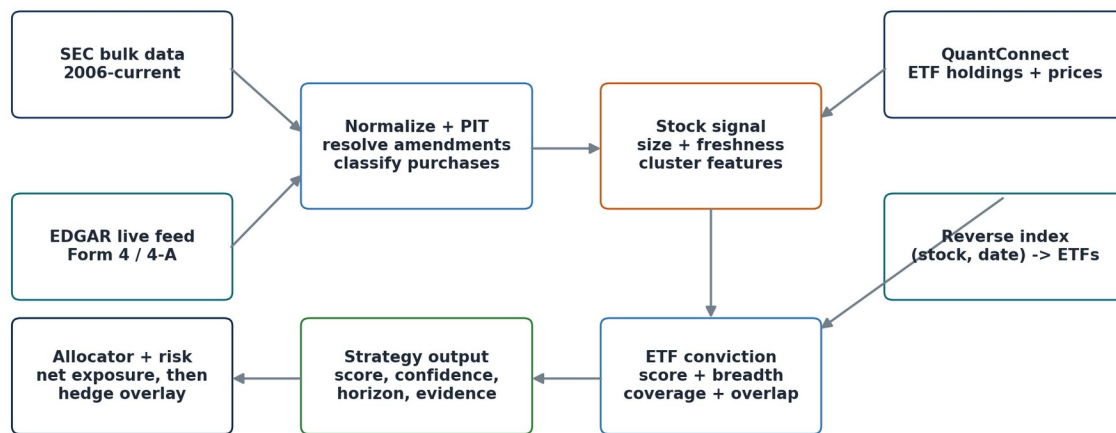
"largest_contribution": 0.36,
"evidence": {
  "unique_companies": 4,
  "unique_buyers": 7,
  "purchase_value_30d": 3250000
},
"risk_flags": ["holdings_snapshot_lag_2d"]
}

```

14.3 Integration with other strategies and hedging

The allocator combines independent signals and nets positions before hedging. For example, the insider strategy may be bullish on semiconductors while a valuation strategy is neutral and a short-interest strategy is bearish. The allocator determines the final exposure; the hedge engine then manages residual market, sector, factor, volatility, and drawdown risk. Hedging this strategy independently would create unnecessary turnover and potentially offset natural diversification between engines.

End-to-End Architecture



Collection -> normalization -> stock evidence -> ETF discovery -> portfolio-level risk control

PART V

Research Execution

Test the information in the correct order and reject attractive leakage.



15. Backtest Sequence and Controls

The research must first establish that the filings predict stock returns after disclosure. Only then should it test whether ETF aggregation improves monetization. Skipping the stock event study makes it difficult to diagnose whether a weak ETF result comes from the raw signal, the mapping, or the portfolio rules.

Round	Objective	Deliverable
R0	Data integrity	Ingestion, timestamps, amendments, mapping, coverage; no performance results.
R1	Stock event study	Code-P purchase events; 1d, 5d, 20d, 60d excess returns.
R2	Size and freshness	Purchase value and decay half-lives; out-of-sample comparison.
R3	Within-company clustering	Unique buyers, repeated dates, role breadth.
R4	ETF discovery baseline	Seed count and weighted exposure using PIT holdings.
R5	ETF conviction	Weighted standardized stock scores, breadth, concentration.
R6	Confounder controls	Momentum, drawdown, valuation, earnings, size, liquidity, sector.
R7	Portfolio simulation	Weekly top-N portfolio, costs, overlap, turnover, capacity.
R8	Allocator contribution	Correlation and marginal portfolio value versus other alphas.
R9	Advanced features	10b5-1, opportunistic/routine, insider quality, ML.

15.1 Stock event-study design

- Event time zero is the first tradable time after public filing availability.
- Calculate raw, market-adjusted, sector-adjusted, and factor-adjusted returns.
- Use event windows of 1, 5, 10, 20, 40, 60, and 120 trading days; primary windows are 5, 20, and 60 days.
- Cluster standard errors by issuer and date, or use a two-way clustering method appropriate to the panel.
- Avoid overlapping-event inflation by testing one event per issuer/day and by constructing non-overlapping portfolio variants.
- Compare purchases with matched non-event firms by size, industry, momentum, valuation, and liquidity.

15.2 ETF cross-sectional tests

- Rank eligible candidate ETFs by each score family at every rebalance date.
- Report quintiles/deciles, top-minus-bottom spreads, monotonicity, and information coefficient.
- Compare stock-first candidate selection with an all-ETF scoring baseline to measure any selection effect.
- Compare ETF aggregation against direct equal-weight ownership of the seed stocks to determine whether diversification helps or merely dilutes the signal.
- Compare against ordinary ETF momentum, sector momentum, equal-weight eligible ETFs, and random within-sector assignments.

15.3 Essential confounders

Confounder	Why it matters	Required test
Prior return / momentum	Insiders may buy after declines or into improving trends	Control 1m, 3m, 6m, 12-1 momentum and drawdown.
Value	Purchases may proxy undervaluation	Book-to-market, earnings yield, FCF yield where PIT data exists.
Size and liquidity	Signal may concentrate in small illiquid firms	Market cap, ADV, spread, idiosyncratic volatility.
Earnings events	Purchases can occur near earnings or guidance changes	Exclude +/-2 and +/-5 trading days; compare.
Buybacks and financing	Corporate repurchases or offerings may affect returns	Tag material capital-structure events where feasible.
Sector/theme momentum	ETF results may be ordinary industry continuation	Sector benchmark and ETF momentum controls.
Market regime	Signal may work only in risk-on or recovery periods	Bull/bear trend, VIX/realized volatility, credit conditions.

15.4 Transaction costs and execution

- Use next-open execution as the conservative default; test next-close and one-day-delayed alternatives.

- Run at least 0, 5, 10, and 20 basis points per side or an empirically estimated spread/slippage model.
- Use actual ETF prices and liquidity available on the trade date.
- Report turnover, average holding period, trade count, capacity proxy, and performance sensitivity to execution delay.
- Do not treat the SEC quarterly publication date as signal time; the underlying filing was public at its original EDGAR availability time.

16. Statistics, Robustness, and Stage Gates

Research Stage Gates



No leverage, live deployment, or hedge coupling until the preceding gate passes.

16.1 Required statistics

Category	Required outputs
Signal diagnostics	Observation count, issuer count, unique buyer count, coverage, missingness, score distribution, concentration.
Cross-sectional	Mean/median IC, rank IC, ICIR, positive-period rate, quintile monotonicity, top-minus-bottom spread.
Portfolio	CAGR, annualized volatility, Sharpe, Sortino, max drawdown, Calmar, expected shortfall, beta, alpha, turnover.
Reliability	HAC/two-way clustered t-statistics, block-bootstrap confidence intervals, year-by-year and regime results.
Diversification	Correlation and marginal Sharpe contribution versus the other strategy engines.

16.2 Negative controls and placebo tests

- Shuffle filing dates within issuer while preserving event counts.
- Shuffle purchase events across stocks within sector and size bucket.
- Shuffle stock-to-ETF membership while preserving ETF constituent counts.
- Use grants (code A) as a placebo event family.
- Use stale filings older than the signal window as a placebo.
- Randomly select ETFs from the same sector and liquidity bucket as each candidate ETF.

16.3 Walk-forward and holdout

Example rolling protocol

Train: 2015-2018
 Validate: 2019-2020
 Test: 2021

Train: 2016-2019
 Validate: 2020-2021
 Test: 2022

Continue forward and concatenate test periods only.
 Keep a final untouched period for one-time adjudication.

16.4 Multiple testing

Label every specification PRIMARY, SECONDARY, or EXPLORATORY. Half-lives, thresholds, role definitions, cluster windows, ETF exposure requirements, and execution lags can create hundreds of combinations. Freeze primary specifications in configuration files before results exist; apply false-discovery-rate control to broad batteries; never rename the best exploratory run as the original hypothesis.

16.5 Specification ledger

Frozen family	Fields that must be recorded
Signal definition	Transaction family, security class, role eligibility, minimum value, lookback, decay half-life.
Availability	Timestamp hierarchy, execution delay, amendment treatment, and ETF-holdings lag.
Universe	Stock and ETF eligibility, liquidity, age, coverage, and duplicate-ETF rules.
Portfolio	Rebalance schedule, top-N, weighting, caps, cash rule, cost model, and benchmark.
Evaluation	Primary horizons, factor controls, statistics, bootstrap method, and holdout period.
Status	PRIMARY / SECONDARY / EXPLORATORY, preregistration commit, owner, and disposition.

Promotion rule

An exploratory result may become a new primary hypothesis only after it is written into a fresh preregistration and evaluated on data that played no role in discovering it.

PART VI

Implementation Blueprint

Translate the research protocol into reproducible data jobs and agent-ready artifacts.



17. Collection Jobs, Storage, and Code Layout

17.1 Batch jobs

Job	Responsibility
download_sec_insider_bulk	Fetch missing quarterly ZIPs; verify checksums and inventory.
parse_sec_insider_bulk	Load as-filed tables with explicit schemas and lineage.
enrich_edgar_metadata	Recover acceptance timestamp, XML URL, amendment links, and flags by accession.
normalize_insider_transactions	Classify security, code, role, ownership, and purchase value.
build_security_master	Map issuer CIK and historical tickers to QuantConnect identifiers.
build_stock_signals	Generate daily PIT event features and stock scores.
build_etf_reverse_index	Invert eligible ETF holdings by date and stock.
build_etf_signals	Aggregate stock evidence to ETF score families.
run_insider_etf_experiment	Execute frozen configs and write metrics, plots, audit logs.

17.2 Suggested repository structure

```

data/
  manifests/
  raw/sec/insider_bulk/
  raw/sec/edgar_filings/
  normalized/insider/
  normalized/security_master/
  derived/stock_signals/
  derived/stock_etf_reverse_index/
  derived/etf_signals/

alphas/insider_etf/
  filters.py
  amendments.py
  security_classification.py
  stock_signal.py
  etf_aggregation.py
  universe.py
  diagnostics.py
  schemas.py

scripts/
  download_sec_insider_bulk.py
  enrich_edgar_acceptance_times.py
  build_insider_stock_signals.py
  build_stock_etf_reverse_index.py
  run_insider_etf_experiment.py

tests/
  test_form4_filters.py
  test_amendment_reconciliation.py
  test_no_pre_filing_signal.py
  test_security_mapping_pit.py
  test_etf_holdings_pit.py
  test_joint_filer_dedup.py
  test_weight_sums.py

research/insider_etf/
  preregistration/
  configs/
  experiments/
  reports/

```

17.3 Mandatory automated tests

Test	Invariant
test_no_pre_filing_signal	No event enters a stock score before its public availability time.
test_transaction_date_not_used_as_availability	Transaction date can never drive execution.
test_form4a_supersedes_original	Corrected records are not double-counted.
test_joint_owner_no_value_multiplication	One economic transaction is not multiplied by owner rows.
test_code_p_acquired_only	V1 contains only code P with acquired indicator A.
test_common_equity_classifier	Restricted/derivative/non-common titles are excluded or quarantined.
test_security_mapping_asof	Ticker/CIK/QC mapping is valid on the event date.
test_etf_snapshot_available	ETF holdings used were known by the signal date or lagged conservatively.
test_etf_weight_sum	Snapshot weight totals are plausible and missing weight is measured.
test_reproducible_snapshot	Same config + data hashes + commit reproduce identical results.

18. Worked Example and Pseudocode

18.1 Fictional stock-score example

Issuer	Qualifying purchases	Age (trading days)	Raw stock score	Cluster
Stock A	\$1,200,000	3	2.90	1 buyer
Stock B	\$400,000	8	1.67	1 buyer
Stock C	\$400,000 total	4 and 10	2.54	2 buyers

ETF	Stock A wt.	Stock B wt.	Stock C wt.	Seed count	Seed exposure	Raw weighted score
ETF X	8%	6%	5%	3	19%	0.459
ETF Y	2%	1%	0%	2	3%	0.075

ETF X ranks higher because the evidence spans more companies and represents a much larger share of the fund. In production, stock scores would be standardized cross-sectionally, and concentration/overlap diagnostics would accompany the ranking.

18.2 Reference pseudocode

```
def build_daily_stock_scores(as_of, transactions):
    events = transactions[
        (transactions.available_at <= as_of)
        & (transactions.age_trading_days <= 30)
        & (transactions.document_type == "4")
        & (transactions.transaction_code == "P")
        & (transactions.acquired_disposed == "A")
        & (transactions.is_common_equity)
        & (transactions.is_officer_or_director)
        & (transactions.purchase_value >= 50_000)
        & (~transactions.is_superseded)
    ]

    events["size"] = log1p(events.purchase_value / 50_000)
    events["freshness"] = exp(-log(2) * events.age_trading_days / 20)
    events["event_score"] = events["size"] * events["freshness"]

    stocks = aggregate_by_security(events,
        raw_score=sum("event_score"),
        unique_buyers=nunique("reporting_owner_id"),
        purchase_value=sum("purchase_value"))

    stocks["score"] = cross_sectional_winsorized_z(stocks.raw_score)
    return stocks
```

```
def discover_and_score_etfs(as_of, stock_scores, reverse_index, eligible_etfs):
    seeds = select_hybrid_seeds(stock_scores)
    candidates = union(reverse_index[(as_of, s)] for s in seeds.security_id)
    candidates = candidates.intersection(eligible_etfs(as_of))

    rows = join_candidate_holdings(candidates, seeds, as_of)
    return aggregate_by_etf(rows,
        seed_count=nunique("security_id"),
        seed_exposure=sum("etf_weight"),
        conviction=sum("etf_weight * stock_score"),
        largest_contribution=max("etf_weight * stock_score"))
```

19. Go/No-Go Criteria and Risk Register

19.1 Data go/no-go

Disposition	Condition
GO	At least 99% of candidate ETF weight maps to valid securities; exact or conservative filing availability exists; amendments and joint filers are tested; ETF snapshot lag is documented.
REMEDiate	Coverage is adequate but one field family is weak, such as ownership increase or 10b5-1. Continue with a simpler score and mark the missing feature unavailable.
NO-GO	Signal uses transaction dates, current tickers, future ETF holdings, unreconciled amendments, or duplicated joint-owner transactions.

19.2 Signal acceptance gate

- Stock-event returns are positive at one or more primary horizons out of sample and remain economically meaningful after controls.
- Within-company cluster features improve or sharpen the signal without relying on one year or one microcap segment.
- ETF top-minus-bottom spread and information coefficient are positive out of sample, preferably with approximate monotonicity.
- ETF aggregation adds risk-adjusted value relative to direct seed-stock ownership or provides materially lower drawdown/idiosyncratic risk.
- Results survive conservative execution, fees, ETF-overlap controls, and a holdings-availability lag.
- The strategy adds marginal portfolio value when combined with the existing alpha engines; it is not merely another momentum proxy.

19.3 Risk register

Risk	Severity	Mitigation
Look-ahead from transaction date	Critical	Use filing acceptance; next-open fallback.
Amendment double counting	Critical	Version/supersede by accession lineage.
Joint-owner multiplication	Critical	Economic-event dedupe and owner-set modeling.
ETF holdings availability	Critical	Use LastUpdate/EndTime or five-day lag.
Ticker/security mismatch	High	CIK + historical security master.
Value outliers/private placements	High	Log scale, winsorization, separate robustness.
Microcap/illiquidity concentration	High	Stock context controls and ETF liquidity filter.
Selection/p-hacking	High	Pre-registration, holdout, FDR, immutable experiments.
Weak bearish counterpart	Medium	Keep V1 long-only; research sales separately.
Crowded ETF duplicates	Medium	Overlap clustering and exposure caps.
Vendor/data revisions	Medium	Immutable SEC snapshots and hashes.

Canonical V1 in one paragraph

From SEC Form 4/4-A data, identify non-derivative common-stock code-P acquisitions by officers or directors with valid price and shares and at least \$50,000 of purchase value. Timestamp them by public filing availability, aggregate 30 days of events with a 20-trading-day half-life and logarithmic value scaling, standardize stock scores, select high-conviction seeds, invert point-in-time eligible ETF holdings, and rank ETFs by the sum of constituent weight times stock score. Require at least two seed companies and 5% seed exposure, rebalance weekly into the top 3-5 long ETFs, apply no leverage, and adjudicate with walk-forward out-of-sample tests and conservative costs.

19.4 First implementation milestone

Step	Definition of done
1	Download and hash all SEC quarterly insider packages from 2006 through the latest complete quarter.
2	Parse the eight linked tables with explicit schemas and build accession-level lineage.
3	Enrich a representative sample with exact EDGAR acceptance times and reconcile Form 4/A amendments.
4	Create the common-equity/code-P/officer-or-director V1 filter and a quarantine report.
5	Map issuers to point-in-time market securities; quantify mapped portfolio weight rather than ticker count alone.
6	Run the stock-level event study before constructing an ETF portfolio.
7	Build and audit the historical stock-to-ETF reverse index, including conservative holdings lags.
8	Freeze R4 and R5 preregistrations, then execute ETF ranking and portfolio tests.

The milestone is complete only when the data-audit report can trace every scored event from experiment output back to its SEC accession, owner rows, transaction row, security mapping, and ETF snapshot.

APPENDICES

Reference Tables and Research Artifacts

Operational details for implementation, review, and reproducibility.



Appendix A. Form-Code Reference

Code	Meaning	Treatment
P	Open-market or private purchase of non-derivative or derivative security	Primary purchase code
S	Open-market or private sale	Separate future strategy
V	Voluntarily reported earlier than required	Context flag
A	Grant, award, or other Rule 16b-3 acquisition	Exclude V1
D	Disposition to issuer under Rule 16b-3	Exclude V1
F	Payment of exercise price/tax liability by withholding securities	Exclude V1
I	Discretionary Rule 16b-3 transaction	Exclude V1
M	Exercise or conversion of derivative security	Exclude V1
C/E/H/O/X	Derivative conversion/expiration/exercise codes	Exclude V1
G	Bona fide gift	Exclude V1
L	Small acquisition under Rule 16a-6	Separate research
W/Z	Will/descent or voting-trust transaction	Exclude V1
J	Other acquisition/disposition described in footnote	Quarantine
K	Equity swap or similar instrument	Exclude V1
U	Disposition in change-of-control tender	Exclude V1

Appendix B. Minimum Field Dictionary

Field	Definition	Use
accession_number	SEC immutable filing identifier	Raw / normalized
acceptance_timestamp	Public availability time	PIT execution
filing_date / transaction_date	Disclosure date / insider action date	Audit and lag
issuer_cik / security_id	Entity and tradable security keys	Cross-source join
reporting_owner_cik	Insider identity	Unique buyer/role history
role_flags / title	Officer, director, owner, title	Eligibility/features
security_title_raw / class	As-filed and normalized security	Common-equity filter
transaction_code	P, A, M, etc.	Event meaning
acquired_disposed_code	A or D	Purchase direction
shares / price / value	Economic size	Signal input
shares_owned_after	Post-event ownership	Ownership increase
direct_indirect	Ownership channel	Primary/secondary sample
is_10b5_1	Plan indicator where available	Discretion feature
is_amendment / superseded	Version status	Dedupe/correction
footnote_ids	Interpretation references	Exception handling
source_snapshot_version	Immutable dataset lineage	Reproducibility

Appendix C. Experiment IDs and Artifact Contract

Experiment	Purpose
INSETF-R0-DATA	PIT/data audit only

Experiment	Purpose
INSETF-R1-STOCK-P	Code-P stock event study
INSETF-R2-DECAY05/10/20/40/60	Freshness half-life battery
INSETF-R3-CLUSTER	Unique-buyer and date breadth
INSETF-R4-ETF-COUNT	Seed-count ETF baseline
INSETF-R5-ETF-WEIGHT	Weighted exposure and conviction
INSETF-R6-CONTROLS	Momentum/value/earnings/liquidity controls
INSETF-R7-PORTFOLIO	Investable weekly long-only simulation
INSETF-R8-ALLOCATOR	Marginal multi-strategy contribution
INSETF-R9-ADVANCED	10b5-1, opportunistic/routine, ML

Every experiment must persist:

- experiment_id
- preregistration_commit
- code_commit
- config_hash
- data_snapshot_hashes
- security_master_version
- ETF_holdings_version
- research / validation / test dates
- execution assumptions
- cost model
- PIT audit result
- raw metrics and plots
- warnings and invalidation reason
- disposition: PASS / FAIL / INCONCLUSIVE / INVALID-DATA

Appendix D. AI-Engine Instructions

- Inspect the existing repository and reuse backtest, PIT universe, security master, metrics, and risk components rather than duplicating them.
- Implement Round 0 first. Do not run performance research until the data audit passes.
- Keep raw collection, normalization, feature construction, ETF aggregation, portfolio allocation, and hedging in separate modules.
- Do not call SEC or vendor APIs dynamically inside historical backtests.
- Pre-register every experiment and commit the configuration before results are generated.
- Do not suppress ambiguous transactions; quarantine them with explicit reason codes.
- Write deterministic tests for availability, amendments, joint owners, security mapping, and ETF holdings lag.
- Keep insider sales, derivative activity, leverage, options, and machine learning outside canonical V1.
- Produce a disposition for every experiment: PASS, FAIL, INCONCLUSIVE, or INVALID-DATA.
- Stop escalation when a gate fails. A more complex model is not a remedy for a contaminated dataset.

Appendix E. Sources and Further Reading

The strategy design is original research work. The following sources support regulatory facts, data availability, implementation interfaces, and adjacent empirical motivation. Accessed August 2026.

- SEC - Insider Transactions Data Sets** - Free quarterly flattened Forms 3, 4, and 5 data; current page coverage and downloads. [Source](#)
- SEC - Insider Transactions Data Sets documentation** - Scope, eight-table organization, file format, keys, and field definitions. [Source](#)
- SEC - Ownership Form Codes** - Official transaction-code definitions, including P, S, A, F, M, G, and J. [Source](#)
- Investor.gov - Insider Transactions and Forms 3, 4, and 5** - Form 4 timing and investor explanation of transaction reporting. [Source](#)
- SEC - Developer Resources** - EDGAR access, archives, and fair-access guidance. [Source](#)
- Financial Modeling Prep - Latest Insider Trading API** - Optional normalized feed with transaction dates, roles, details, and filing links. [Source](#)
- QuantConnect - US ETF Constituents** - Historical coverage, frequency, weights, security-master dependency, and stated potential publication delay. [Source](#)

[8] **QuantConnect - ETF Constituents Universes** - Historical constituent requests and constituent weight access in LEAN. [Source](#)

[9] **Lakonishok and Lee - Are Insiders' Trades Informative?** - Classic empirical work on insider activity and subsequent returns. [Source](#)

[10] **Cohen, Malloy, and Pomorski - Decoding Inside Information** - Research motivation for distinguishing opportunistic from routine insider trades. [Source](#)

[11] **Do Insiders Cluster Trades With Colleagues?** - Adjacent evidence on clustered insider purchases and subsequent returns. [Source](#)

Final Perspective

This project is attractive because it combines a free, auditable regulatory dataset with a distinctive ETF aggregation hypothesis. Its success should not be judged only by raw CAGR. A modest but persistent signal that improves ETF selection, reduces idiosyncratic exposure, and adds low-correlation evidence to the broader trading agent would be a strong outcome. The research becomes genuinely interesting when two independent forms of clustering survive: multiple insiders buying within issuers, and multiple such issuers concentrating inside the same ETF.

The correct first milestone is therefore not a backtest chart. It is an immutable point-in-time evidence ledger in which every transaction can be traced to its accession, every correction is reconciled, every stock maps to the correct historical security, and every ETF weight was actually knowable on the signal date.